

KSP Datathon 2026 — Challenge 1

Intelligent Conversational AI & Crime Analytics Platform

TRUST DESIGN DECISIONS

Architecture Reference — Component C14 · C15 · C16

Document scope	Three pre-build architectural decisions that determine citation, transparency, and RBAC enforcement
Components	C14 (Citation/Provenance) · C15 (RBAC Policy Engine) · C16 (Audit & Monitoring)
Applies to	All 23 user stories — trust layer is cross-cutting, not a feature module
Decision status	All three decisions finalised here — build accordingly
Related documents	Semantic Layer C5 · Demo Script v1 · Data Architecture v1
Version	v1.0

1. Why Trust Decisions Come Before Architecture

Trust is not a feature checklist to be added after the system is built. It is a set of design decisions that affect the data layer, API shape, and UI layout simultaneously. Making them late means retrofitting — which in a law enforcement context means the system is not deployable even if it works.

Three decisions must be locked before any frontend work begins:

Decision	Stakes if deferred
D1 — How citation works	If citation is retrofitted, the UI has no natural place for it and PDFs omit source trails — the remand-hearing use case (Story 6) breaks
D2 — What 'I don't know' looks like	If undefined, the LLM fills uncertainty with confident-sounding hallucination — catastrophic in a policing context
D3 — Where RBAC is enforced	If enforced at the prompt layer instead of the data layer, a clever query bypasses it — the system is not actually secure

2. Decision 1 — How Citation Works (C14)

An LLM answer that says 'Rajan Gowda has 3 prior cases' is worse than useless in a remand hearing if the IO cannot immediately point to which FIRs. The citation layer (C14) converts a plausible-sounding answer into a legally actionable one. Peregrine's design — clickable citations down to the row in the source record — is the benchmark.

The Citation Unit — Three Levels

Citation level	What it looks like	Triggered by	Risk if omitted
FIR-level	Source: KA-BLR-047-2021-0183	Aggregate / count queries	Sufficient for counts; insufficient for individual factual claims
Field-level	Source: KA-BLR-047-2021-0183 · fir_accused[A1].is_arrested	Lookup queries about a specific person or case	Evaluator cannot verify the specific fact being asserted
Sentence-level (RAG)	Source: KA-BLR-047-2021-0183 · case_diary_entry #4 · 2023-08-11	MO match, case brief — answers drawing on narrative free text	Hallucination is undetectable without sentence-level sourcing

DECISION Implement all three levels, triggered automatically by query type. The query orchestrator (C4) already knows which engine was used — SQL queries produce FIR-level or field-level citations; RAG queries produce sentence-level citations. The citation payload is a structured JSON object attached to every response.

Citation Response Object — Required Schema

```
{
  "answer": "Rajan Gowda has 3 prior cases in Karnataka across 2 stations.",
  "citations": [
    {
      "claim": "3 prior cases in Karnataka",
      "source_type": "sql_aggregate",
      "records": [
        { "table": "fir_accused", "fir_id": "KA-BLR-047-2021-0183", "field": "person_id" },
        { "table": "fir_accused", "fir_id": "KA-MYS-012-2022-0441", "field": "person_id" },
        { "table": "fir_accused", "fir_id": "KA-BLR-031-2023-0067", "field": "person_id" }
      ]
    },
    {
      "claim": "across 2 stations",
      "source_type": "sql_aggregate",
      "records": [
        { "table": "police_stations", "station_id": "KA-BLR-047" },
        { "table": "police_stations", "station_id": "KA-MYS-012" }
      ]
    }
  ],
  "generated_sql": "SELECT f.fir_id, f.station_id FROM firs f JOIN fir_accused fa ON f.fir_id = fa.fir_id WHERE fa.person_id = 'uuid-xxx'",
  "er_confidence": 0.91,
  "confidence": "high",
  "query_type": "person_lookup"
}
```

UI Consequences — Four Binding Decisions

UI element	Decision
Inline citations	Citation numbers [1] [2] are rendered as superscripts inline in every answer bubble — not in a footnote. Every factual claim gets one.
Source panel	Right sidebar (not a modal) expands when a citation is clicked — shows the actual record row with the cited field highlighted. Sidebar avoids breaking conversation flow.
Generated SQL	Shown in a collapsible 'Reasoning' section beneath every SQL-backed answer, collapsed by default. Expanded in at least one demo beat (Beat 3 recommended).

UI element	Decision
PDF export (Story 6)	PDF must include the full citation list as a numbered footnote section. Not optional — this is what makes the PDF legally usable at a remand hearing.
Always-on principle	Citations are NEVER a 'trust callout' feature shown only in one demo segment. They appear in every beat, every response, from the first query onward.

3. Decision 2 — What 'I Don't Know' Looks Like (C14 + C6/C7)

An LLM that fills data gaps with confident-sounding answers is worse than no system at all in a policing context. 'The accused has no prior history in Karnataka' stated confidently when the real answer is 'data for 3 districts may be incomplete' could affect a remand decision. The system must have a designed, deliberate uncertainty state.

DESIGN RULE *The system never says 'I don't know' and stops. It always states what it does know, what the limit is, and what the user can do next. This transforms a dead end into a useful interaction — and answers Module 9 (explainable AI) without a separate UI panel.*

Three Distinct Uncertainty Situations

Situation A — Data Genuinely Does Not Exist

Query: 'Does Mohan Reddy have prior cases in Karnataka?'

No records found for 'Mohan Reddy' in the Karnataka crime database after matching across name variants (3 variants checked).

This means either:

- (a) This person has no registered cases in Karnataka, or
- (b) Any existing records may be under an unmatched name variant.

Suggestion: retry with father's name or date of birth if available.

[Source: searched fir_accused, persons – 0 matches]
[er_confidence: N/A – no candidate records found]

Situation B — Data Exists but Quality Is Insufficient

Query: 'Show me a hotspot map of vehicle thefts in Mysuru this year.'

Mysuru district has 412 vehicle-theft FIRs in the requested period. However, 38% of these FIRs (156 records) are missing location coordinates.

A geographic analysis on this data would be unreliable.

I can show you instead:

- (a) A breakdown by station name – covers 100% of FIRs, or
- (b) The 256 FIRs that do have coordinates, with a stated caveat.

Which would be more useful?

[Data quality: 62% coordinate coverage, Mysuru, Jan-Jun 2025]
[Confidence: LOW – geographic analysis not recommended on this dataset]

Situation C — Query Is Out of Scope for the User's Role

Query (IO at Indiranagar PS): 'Show me victim addresses for all chain-snatching cases in Bengaluru Urban.'

This query requires access to victim personal details across all stations in Bengaluru Urban district.

Your current role (Inspector · Indiranagar PS) gives you access to victim details only for cases registered at Indiranagar PS.

If you need cross-station victim information for a specific investigation, contact your Circle Inspector or DCRB Analyst for an authorised data pull.

[Access level: station-scoped · Role: IO · Records withheld: victim.address]

Implementation Decisions

Requirement	Decision
Confidence field	Every response object carries a confidence field: high / medium / low / insufficient. This is computed before the LLM generates text, not after.
UI indicator	low and insufficient confidence trigger a visible yellow or red badge beside the answer bubble — not buried in a footnote. The badge is always visible, not expandable-only.
Data quality pre-check	A data_quality_flags query runs BEFORE the LLM sees the question. Result is injected into the prompt context. Prevents the LLM from generating confident text that the citation layer then undercuts.
Uncertainty logged	'I don't know' responses are logged in query_audit_log identically to successful answers. They are as important for understanding system behaviour as successes.
No silent failure	If the SQL query returns 0 rows, the system returns Situation A explicitly — never an empty chat bubble.
No hallucination fallback	If the LLM cannot ground an answer in retrieved records, it must return Situation A or B — never generate an answer from training data. This is enforced by the grounding validator (Step 4 of C5 build).

4. Decision 3 — Where RBAC Is Enforced (C15)

RBAC at the prompt layer — telling the LLM 'you are an IO, don't show other stations' victim data' — is not enforcement. It is a suggestion. A cleverly phrased query, an edge case the prompt didn't anticipate, or a multi-turn session that drifts context can bypass it. Role scope must be enforced at the data layer.

DECISION RBAC is enforced at the database view layer — not at the application layer and not at the prompt layer. The LLM generates SQL against a role-scoped view, not the base table. The API connection has *SELECT* permission only on views, not on base tables. This makes bypass structurally impossible, not just policy-forbidden.

The Two-Layer Enforcement Model

Layer 1 — Query Construction (C5 + C6 + C4)

Before the LLM generates SQL, the session context payload (from SC5 of the semantic layer) injects mandatory WHERE clauses into the SQL generation prompt. These are not instructions to the LLM — they are constraints on the SQL template that the LLM cannot override.

```
-- Example: IO at Indiranagar PS (KA-BLR-047)
-- Mandatory base filter injected for every SQL query:

WHERE firs.station_id = 'KA-BLR-047'
  AND firs.io_officer_id = 'KSP-12345'    -- for 'my cases' queries

-- Exception: antecedent lookup on named person (Story 1)
--   Base filter overridden to cross-district read
--   Response scoped to FIR header only (no case diary, no victim PII)

-- Example: SP Meena, Mysuru district
WHERE firs.district_id = 'MYS'
  AND result_columns EXCLUDE (victim_name, victim_address, victim_phone)
```

Layer 2 — Result Filtering (C15 applied post-execution)

Even if a generated SQL query somehow returns rows it should not, a post-execution filter strips any field the role is not permitted to see before the result reaches the LLM for response generation. The LLM never sees restricted data — it is not being asked to self-censor.

Role	Fields stripped before LLM sees result	Passes
IO / SHO	fir_victims.address_current · fir_victims.phone_primary (for non-linked cases) · persons.aadhaar_last4 (non-linked) · persons.religion (individual) · persons.caste (individual)	Full case detail for own station / assigned cases
Supervisor	All victim PII · Individual person addresses · persons.religion and caste at individual level	Aggregated trends · Case summaries · Station comparisons
Analyst	All victim PII · All individual person identity fields	Aggregated tables · Pattern data · Cluster results
Policymaker	All individual records entirely — no FIR IDs, no person names, no addresses	Aggregated district/state statistics only
Admin	Nothing stripped — full access for audit purposes	All data + all audit logs

Database View Architecture

One view per role — pre-baked WHERE clauses and column exclusions at the database level. The LLM SQL generator targets the view, never the base table.

```
-- Role-scoped views (created once at DB setup)
CREATE VIEW firs_io_view AS
  SELECT fir_id, station_id, crime_type_id, registration_date,
         investigation_status, chargesheet_deadline,
         io_officer_id, primary_bns_section
  FROM firs;
-- Row-level security applied at query time via session variable

CREATE VIEW firs_supervisor_view AS
  SELECT fir_id, station_id, district_id, crime_type_id,
         registration_date, investigation_status
  FROM firs;
-- No victim PII columns included

CREATE VIEW firs_policymaker_view AS
  SELECT district_id, crime_type_id,
         DATE_TRUNC('month', registration_date) AS month,
         COUNT(*) AS case_count
  FROM firs
  GROUP BY district_id, crime_type_id, month;
-- Only pre-aggregated – no individual records

-- API DB connection: GRANT SELECT ON *_view TO api_user;
-- REVOKE SELECT ON firs, persons, fir_victims FROM api_user;
```

Cross-District Antecedent Exception

The antecedent lookup (Story 1) is the one legitimate case where an IO reads data from outside their station. The resolver handles it as a controlled exception — not a prompt instruction.

Rule	Detail
What is permitted	IO can read fir_accused.person_id across all stations statewide for a named person antecedent lookup
What the response shows	FIR ID, crime type, section, station name, investigation status — nothing more from other stations
What is withheld	Case diary entries, victim details, witness statements, full FIR narrative from stations other than the IO's own
Audit treatment	Query logged normally. records_accessed array is fully populated. No flag raised for legitimate antecedent lookup.
How it is triggered	Query orchestrator (C4) classifies intent as 'person_antecedent_lookup' → applies the exception view rather than the standard IO view

5. Summary — The Three Decisions at a Glance

Decision	What was decided	Component	Build consequence
D1 – Citation	Three-level citation (FIR / field / sentence) attached to every response as a structured JSON payload. Inline [1][2] superscripts in the UI. Right-sidebar source panel. Generated SQL always collapsible in responses.	C14	Citation schema must be designed before API response contract is finalised. PDF export must include footnotes.
D2 – Uncertainty	Three typed uncertainty states (no data / low quality / out of scope). Confidence field on every response. Data quality pre-check runs before LLM generation. Yellow/red badge in UI. Never silent failure.	C14 + C6/C7	Data quality view must be built before query pipeline. Confidence badge must be a UI primitive, not an afterthought.
D3 – RBAC	Enforcement at the database view layer. One view per role. API connection has SELECT only on views. Post-execution column strip before LLM sees results. Cross-district antecedent is the only named exception.	C15	Database views must be the first DB artifact created, before any SQL generation prompt is written. No base-table access for API user.

IMPLEMENTATION ORDER RBAC views (D3) → Citation schema (D1) → Uncertainty states and data quality pre-check (D2) → SQL generation prompt (C6) → UI citation rendering. This order means the trust layer is foundational, not retrofitted.

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